

ESM_2 The Survey Questions

The Survey Questions

(From codebook included in ESM_1)

Compare corresponding survey questions between RDS and NSUM methods:

- Q70/Q71 (document withholding)
- Q39+Q42/Q43 (pay issues)
- Q45+Q47+Q48/Q49 (abuse/threats)
- Q61+Q62/Q64 (excessive hours)
- Q78/Q79 (access to help)

Risk Index Implementation

- Clean coding for 13 risk categories with proper weightings
- NRM referral (0.35)
- Forced labor indicators (0.55 total)
- Below minimum wage (0.10)

Data Processing

The process of coding variables to make them comparable across the Respondent-Driven Sampling (RDS) and Network Scale-Up Method (NSUM) estimations is a multi-step procedure that moves from conceptual mapping to specific data transformations. The core challenge is bridging the “egocentric” questions (about the respondent’s own experiences, used for RDS) with the network-based questions (about how many others the respondent knows with certain experiences, used for NSUM). The fundamental challenge is to create a fair comparison between the “egocentric” RDS questions (about personal experience) and the broader “how many do you know” NSUM questions. This was accomplished by first identifying thematic links between questions and then implementing a specific coding scheme to transform them into a consistent format.

Thematic Groups for Comparison:

- Document Withholding (Most Confident): Comparing personal experience (Q70) with knowing others who lack access to their documents (Q71).
- Pay/Debt Issues (High Confidence): Linking personal experience with debt (Q39) and withheld pay (Q42) to knowing others with such problems (Q43).
- Threats/Abuse (High Confidence): Aggregating personal experiences of being forced, threatened, intimidated, or verbally abused (Q45, Q47, Q48) and comparing them to knowing others who experienced threats or force (Q49).

- **Excessive Hours (Lower Confidence):** Combining questions about inadequate weekly rest (Q61) and excessive overtime (Q62) to compare with knowing others who face similar labor rights issues (Q64). The confidence is lower because the NSUM question (Q64) also asks about annual leave, which is not covered in the corresponding RDS questions.
- **Access to Help (Least Confident):** Matching whether the respondent personally knows where to go for help (Q78) with knowing others who do not (Q79). Coding and Transformation Process The conceptual mapping was then put into practice through a detailed coding process, which is explicitly documented in the project's R script.

The primary goal was to convert all variables into a binary (0/1) format, representing “No” or “Yes” to experiencing or knowing someone who experienced the issue.

Key Coding Steps:

1. **Handling Ambiguous Answers:** For most RDS questions, responses like “I don’t know” or “Prefer not to say” were recoded as NA (missing) to exclude them from the binary analysis.
2. **Converting Categorical RDS Variables to Binary:** Questions with multiple response options (e.g., “Always,” “Often,” “Sometimes,” “Never”) were recoded into a 0/1 format. For instance, for Q70 (document withholding), the answers “Always,” “Often,” and “Sometimes” were all coded as 1 (indicating the presence of the issue), while “Never” was coded as 0.
3. **Aggregating Multiple RDS Questions:** For themes covered by multiple egocentric questions (like “Threats/Abuse”), each individual question was first recoded into a binary 0/1 variable. These binary variables were then combined using a logical OR (|). This means if a respondent answered “Yes” to any of the grouped questions (e.g., experiencing threats OR intimidation OR verbal abuse), the final composite RDS variable would be coded as 1. This aggregation was necessary to match the broader scope of the single corresponding NSUM question.
4. **Converting NSUM Count Variables to Binary:** The NSUM questions, which asked for a numerical count (e.g., “how many do you know?”), were also converted to binary. If a respondent knew one or more people (> 0), their answer was coded as 1; if they knew zero, it remained 0.

Indicators

Creating comparable variables to bridge the Respondent-Driven Sampling (RDS) and Network Scale-Up Method (NSUM) estimations is a key methodological challenge the research team is actively working to solve. The core issue is that the two methods rely on differently framed questions:

- RDS estimation uses “egocentric” questions, which focus on the respondent’s personal experiences (e.g., “Have you been forced to work?”).
- NSUM estimation uses questions about the respondent’s knowledge of others in their network (e.g., “How many domestic workers do you know who have experienced...”). The survey questions designed for these two purposes do not always match up directly, making a fair comparison between the estimates difficult.

The goal is to aggregate some of the specific “egocentric” questions to make them comparable to the broader “how many do you know” NSUM questions. To accomplish this, several sets of “linking” questions have been identified with varying degrees of confidence in their comparability:

- Document Withholding (High Confidence):
 - RDS Question: Q70 asks if the employer has withheld the respondent’s travel and identity documents.
 - NSUM Question: Q71 asks how many other domestic workers the respondent knows who do not have access to their own documents.
- Debt and Pay Issues (High Confidence):
 - RDS Questions: This involves combining Q39 (having to pay off debt to someone who helped find work) and Q42 (has your pay ever been withheld?).
 - NSUM Question: Q43 asks how many others the respondent knows who have experienced problems with debt or pay.
- Abuse and Threats (Medium Confidence):
 - RDS Questions: This requires grouping Q45 (forced, deceived, or threatened into poor working conditions), Q47 (employer intimidation or threats), and Q48 (verbal abuse).
 - NSUM Question: Q49 asks how many others the respondent knows who have experienced the use of threat or force.
- Excessive Hours (Lower Confidence):
 - RDS Questions: Q61 (weekly rest does not last 24 hours) and Q62 (working excessive overtime) are combined.
 - NSUM Question: Q64 asks about knowing others who work excessive overtime, lack breaks, or lack annual leave. The comparison here is considered “flakier” because the RDS questions do not ask about annual leave, creating a partial mismatch.
- Access to Help (Lower Confidence):
 - RDS Question: Q78 asks if the respondent knows who might help them (coded for “No” answers).
 - NSUM Question: Q79 asks how many other domestic workers the respondent knows who do NOT know where to go for help.

ESM_2 Table1: Comprehensive RDS Estimates: Prevalence estimates for all indicators across population sizes and RDS methods (RDS-I, RDS-SS) with neighbourhood bootstrap confidences intervals

Indicator	RDS-I Estimator				RDS-SS Estimator			
	50K	100K	980K	1.74M	50K	100K	980K	1.74M
Document withholding	49.4% (25.5-56.2)	49.4% (25.2-55.9)	49.4% (25.3-56.3)	49.4% (25.4-56.1)	34.2% (25.6-56.2)	34.2% (25.1-56.5)	34.2% (25.2-56.0)	34.2% (25.3-56.5)
Excessive working hours	92.8% (71.8-92.8)	92.8% (71.8-92.5)	92.8% (71.4-92.9)	92.8% (71.7-92.8)	83.7% (71.5-93.0)	83.7% (71.5-93.0)	83.7% (71.5-93.0)	83.7% (71.6-92.8)
Limited access to help	57.7% (54.7-80.0)	57.7% (54.8-79.8)	57.7% (55.1-80.3)	57.7% (54.9-80.0)	59.4% (54.9-80.1)	59.4% (55.1-80.2)	59.4% (54.7-80.2)	59.4% (54.8-80.1)
Pay-related issues	59.3% (34.5-63.5)	59.3% (34.8-63.0)	59.3% (34.2-63.5)	59.3% (34.2-63.6)	49.7% (34.0-63.8)	49.7% (34.2-64.1)	49.7% (34.8-63.4)	49.7% (34.6-63.5)
Threats and abuse	65.2% (40.3-71.7)	65.2% (40.2-72.0)	65.2% (40.5-72.1)	65.2% (40.6-72.1)	57.2% (40.9-72.5)	57.2% (40.9-72.0)	57.2% (40.1-72.3)	57.2% (40.6-72.6)

Note: RDS-I: Salganik-Heckathorn estimator; RDS-SS: Gile's Successive Sampling estimator. Population sizes represent assumed UK domestic worker totals: 50K, 100K, 980K, and 1.74M. Estimates shown as prevalence (95% CI) from neighborhood bootstrap with 1,000 resamples.